

OHRID ST PAUL THE APOSTLE, North Macedonia

WMO: 135780

Lat: 41.180N

Lon: 20.742E

Elev: 705

StdP: 93.14

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-6.3	-13.9	1.2	-6.3	-11.3	1.6	-4.3	9.2	3.1	8.3	1.9	3.3	340	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	31.0	19.4	29.3	19.0	28.0	18.6	21.2	27.6	20.4	26.9	19.6	26.2	2.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	15.1	24.9	18.0	14.1	24.1	17.1	13.3	23.3	65.1	27.7	61.8	26.9	59.3	26.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.6	5.7	-12.1	33.9	2.5	1.3	-13.9	34.8	-15.4	35.6	-16.8	36.3	-18.7	37.3		
			-12.6	23.0	2.4	1.1	-14.3	23.8	-15.7	24.4	-17.0	25.0	-18.7	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	1.6	3.0	6.3	10.1	14.7	18.6	20.9	21.1	16.8	12.2	7.6	3.3
	DBStd	7.47	3.73	3.71	3.37	3.28	2.56	2.91	2.23	2.16	2.80	3.19	3.52	3.81
	HDD10.0	928	261	197	121	36	1	0	0	0	0	18	88	207
	HDD18.3	2763	519	430	373	246	115	32	6	4	61	191	323	464
	CDD10.0	1436	0	1	6	39	146	257	338	344	204	86	14	1
	CDD18.3	229	0	0	0	0	2	38	85	89	15	0	0	0
	CDH23.3	2357	0	0	0	2	42	390	878	892	152	2	0	0
	CDH26.7	629	0	0	0	0	2	88	255	261	23	0	0	0
Wind	WSAvg	2.2	2.3	2.5	2.6	2.1	1.9	2.0	2.1	2.0	2.1	2.0	2.1	2.5
Precipitation	PrecAvg	771	73	67	62	71	58	36	30	27	71	92	100	88
	PrecMax	1156	144	167	105	111	112	80	92	84	148	166	187	180
	PrecMin	528	4	19	6	16	22	9	1	4	9	3	25	6
	PrecStd	167	41	36	27	24	27	19	26	18	37	43	41	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	14.0	18.3	22.9	26.7	31.2	33.0	33.0	29.1	23.1	18.2	14.1
		MCWB	7.9	8.6	11.0	13.2	17.1	20.0	19.9	19.9	17.4	16.0	13.6	10.7
	2%	DB	9.8	11.8	15.9	20.1	24.2	29.0	31.0	31.1	26.9	21.1	16.1	12.0
		MCWB	7.1	7.6	9.2	12.4	16.4	19.3	19.3	19.4	17.4	15.2	12.2	9.2
	5%	DB	8.3	10.1	14.0	18.1	22.9	27.2	29.2	29.3	25.0	19.6	14.8	10.7
		MCWB	6.4	6.7	8.5	11.6	15.8	18.7	19.1	19.0	16.9	14.4	11.4	8.5
	10%	DB	7.1	8.8	12.2	16.1	21.1	25.8	28.0	28.0	23.2	18.1	13.1	9.1
		MCWB	5.4	6.2	7.7	10.7	15.1	18.3	18.8	18.6	16.4	13.9	10.4	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	9.4	11.8	14.4	18.7	21.9	22.3	22.5	19.8	17.6	14.4	11.1
		MCDB	10.6	12.5	16.6	19.9	23.7	27.3	28.5	29.1	25.1	20.9	17.2	13.4
	2%	WB	7.7	8.1	10.2	13.2	17.4	20.6	21.2	21.1	18.5	16.2	12.9	9.9
		MCDB	9.1	10.6	14.1	18.6	22.6	26.5	27.8	27.8	24.3	19.7	15.4	11.6
	5%	WB	6.6	7.3	9.2	12.2	16.5	19.6	20.4	20.2	17.7	15.2	11.8	8.8
		MCDB	8.1	9.5	12.9	16.9	21.6	25.8	27.0	27.0	23.2	18.5	14.1	10.5
	10%	WB	5.5	6.4	8.3	11.3	15.6	18.6	19.6	19.4	16.9	14.3	10.7	7.6
		MCDB	6.9	8.5	11.6	15.4	20.3	24.7	26.3	26.2	22.2	17.4	12.9	9.1
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.7	9.9	10.7	12.0	13.3	14.4	14.6	12.2	10.7	9.0	7.8
		MCDBR	8.9	10.3	13.3	13.8	15.2	15.4	16.5	16.6	15.0	12.7	10.5	9.0
		MCWBR	6.5	6.7	7.8	7.5	8.2	7.6	7.8	7.8	7.6	7.7	6.9	6.6
	5% WB	MCDBR	8.2	9.2	12.1	12.9	14.1	14.7	15.3	15.5	13.7	11.6	9.5	8.2
		MCWBR	6.1	6.4	7.4	7.4	8.1	7.7	7.8	7.8	7.4	7.5	6.6	6.3
Clear-Sky Solar Irradiance	taub	0.306	0.331	0.376	0.426	0.422	0.399	0.405	0.418	0.402	0.367	0.333	0.307	
	taud	2.432	2.353	2.265	2.132	2.206	2.311	2.291	2.251	2.283	2.370	2.432	2.441	
	Ebn at Noon	848	875	868	845	857	875	867	844	831	821	805	814	
	Edh at Noon	79	99	121	149	142	128	129	131	118	97	79	72	
All-Sky Solar Radiation	RadAvg	1.76	2.55	3.55	4.59	5.60	6.54	6.85	6.07	4.43	3.14	2.03	1.54	
	RadStd	0.25	0.34	0.48	0.52	0.44	0.49	0.39	0.40	0.39	0.34	0.20	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (21 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	-123	+102	N/A

Nomenclature: See separate page